

Seat no. _____

[34]

No. of Printed Pages: 02

SARDAR PATEL UNIVERSITY
MSc IT (SEMESTER-II) EXAMINATION 2025
PS02EINT51: Advanced Java



Date : 19/4/2025 , Saturday
Time : 10:00 AM to 01:00 PM

Marks: 70

Q.1 Multiple Choice Questions

[08]

- 1 Which method is called only once in the lifecycle of a servlet?**
A) doGet() B) doPost()
C) init() D) service()
- 2 JSP pages are converted into which type of Java component before execution?**
A) Applet B) Servlet
C) Bean D) Listener
- 3 Which of the following is not a core module of Spring?**
A) Spring MVC B) Spring Boot
C) Spring ORM D) Spring Threading
- 4 In Spring, what is the default scope of a bean?**
A) prototype B) session
C) singleton D) request
- 5 Which Hibernate file is responsible for database configuration?**
A) hibernate.cfg.xml B) hibernate-config.xml
C) hibernate.properties D) persistence.xml
- 6 What is Lazy Initialization in Hibernate?**
A) Objects are initialized at the time of mapping
B) Objects are initialized only when they are accessed
C) Objects are initialized in parallel
D) Objects are never initialized
- 7 Which of the following is not a type of Enterprise Java Bean?**
A) Entity Bean B) Stateful Session Bean
C) Servlet Bean D) Message-Driven Bean
- 8 In which container does Enterprise Java Beans (EJB) run?**
A) JSP Container B) Servlet Container
C) EJB Container D) Applet Container

Q.2 Short Question (Attempt Any SEVEN out of NINE)

[14]

- 1 Explain the lifecycle of a Java Servlet with key methods.**
- 2 What are JSP implicit objects? Name any four.**
- 3 Explain Spring MVC architecture in brief.**
- 4 What is Aspect-Oriented Programming (AOP) in Spring?**
- 5 What is Hibernate ORM and why is it used?**
- 6 Differentiate between Lazy and Eager loading in Hibernate**
- 7 What are the types of Enterprise Java Beans (EJB)?**
- 8 Explain session beans in EJB with its type.**
- 9 What is the role of the JDBC framework in Spring?**

(P.T.O.)

- Q. 3**
- [A]** Explain the lifecycle of a Servlet with a diagram and example. **[06]**
- [B]** What are JSP directives, scripting elements, and action elements? Explain with examples. **[06]**
- OR**
- [B]** Discuss the architecture of JSP and its working. **[06]**
- Q. 4**
- [A]** Explain the Spring Framework Architecture with a diagram. **[06]**
- [B]** What are Spring Beans? Explain definition, scope, and lifecycle of beans in Spring. **[06]**
- OR**
- [B]** Explain Spring Security and its importance in web applications. **[06]**
- Q. 5**
- [A]** Explain object relational persistence in Hibernate with an example. **[06]**
- [B]** Discuss different Hibernate mapping techniques with examples. **[06]**
- OR**
- [B]** Explain how Hibernate manages entity identity with an example. **[06]**
- Q. 6**
- [A]** Explain the Java EE architecture and its key components. **[06]**
- [B]** Discuss Message-Driven Beans (MDBs) with an example. **[06]**
- OR**
- [B]** Explain Entity Beans and their role in Java EE applications. **[06]**

—————x—————